

$$\begin{aligned} & \frac{1}{16} s_4^2 s_2^2 (g_1^2 + g_2^2) + \frac{1}{1768} s_4^2 s_2^2 (g_1^2 + g_2^2) (16 g_2^2 + 6 c_2^2) + \frac{1}{9} s_2^2 (g_1^2 + g_2^2) (g_1^2 + g_2^2) + \\ & 9 s_2^2 (g_1^2 + g_2^2) (g_1^2 + 3 g_2^2) + \frac{1}{5} s_2^2 (g_1^2 + g_2^2) (g_1^2 + g_2^2) L F_{5,0}[m_2] + \\ & \frac{1}{12} s_4^2 c_2^2 (g_1^2 + g_2^2) L F_{4,1}[m_2] - \frac{1}{45} s_4^2 c_2^2 (g_1^2 + g_2^2) L F_{5,1}[m_2] + \\ & \frac{1}{128} s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (4 s_2^2 (c_2^2 + c_4 - s_2^2) - s_4 c_2^2) L F_{1,0}[m_1] (g_1^2 + g_2^2)^2 - \\ & \frac{1}{64} s_2^2 s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (g_1^2 + g_2^2) L F_{2,0}[m_1] - \frac{1}{216} s_2^2 s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} L F_{3,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{64} s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (-2 s_2^2 (c_2^2 + c_4 - s_2^2) - s_4 c_2^2) L F_{1,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{16} s_2^2 s_4^2 s_2^2 \frac{1}{m_1^2} \sqrt{p} \sqrt{p} (c_2^2 + 2 c_2^2) (g_1^2 + g_2^2) L F_{2,0}[m_1] + \frac{1}{12} s_4^2 c_2^2 c_2^2 \sqrt{p} \sqrt{p} \\ & L F_{3,0}[m_1] - \frac{1}{128} s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (4 s_2^2 (c_2^2 + c_4 - s_2^2) - s_4 c_2^2) L F_{1,0}[m_1] (g_1^2 + g_2^2)^2 - \\ & \frac{1}{64} s_2^2 s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (g_1^2 + g_2^2) L F_{2,0}[m_1] - \frac{1}{24} s_2^2 s_4^2 c_2^2 L F_{3,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{64} s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (-2 s_2^2 (c_2^2 + c_4 - s_2^2) - s_4 c_2^2) L F_{1,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{16} s_2^2 s_4^4 s_2^2 \frac{1}{m_1^2} \sqrt{p} \sqrt{p} (c_2^2 + 2 c_2^2) (g_1^2 + g_2^2) L F_{2,0}[m_1] + \frac{1}{4} s_4^4 c_2^2 c_2^2 \sqrt{p} \sqrt{p} \\ & \sqrt{p} L F_{3,0}[m_1] + \frac{1}{128} s_4^4 s_2^2 \frac{1}{m_1^2} (2 \sqrt{p} \sqrt{p} \sqrt{p} (-2 s_2^2 (c_2^2 + c_4 - s_2^2) - s_4 c_2^2) - \\ & s_4 s_2^2 (4 s_2^2 (c_2^2 + c_4 - s_2^2) + c_4 c_2^2)) L F_{1,0}[m_1] (g_1^2 + g_2^2)^2 - \\ & \frac{1}{12} s_2^2 s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (\sqrt{p} \sqrt{p} \sqrt{p} (c_2^2 + 2 c_2^2) (g_1^2 + g_2^2) + \sum c_2^2 (g_1^2 + g_2^2)) L F_{2,0}[m_1] + \\ & \frac{1}{288} \sum c_2^2 (3 c_2^2 g_1^2 + g_1^2 g_2^2 + 8 + 9 c_2^2 (1 - 2 c_2^2)) + 2 (g_1^2 - 4 - 9 c_2^2)) L F_{3,0}[m_1] + \\ & \frac{1}{96} s_2^2 s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (-24 c_2^2 c_2^2 \sqrt{p} \sqrt{p} + \sum c_2^2 (-5 - 6 c_2^2)) L F_{4,1}[m_1] + \\ & \frac{1}{45} s_2^2 s_4^2 c_2^2 (g_1^2 + g_2^2) L F_{5,2}[m_1] - \\ & \frac{1}{128} s_4^4 (6 \sqrt{p} \sqrt{p} (-2 s_2^2 (c_2^2 + c_4 - s_2^2) + s_4 c_2^2) - 6 \sqrt{p} \sqrt{p} \sqrt{p} \\ & (2 s_2^2 (c_2^2 + c_4 - s_2^2) + s_4 c_2^2) + \sum c_2^2 (4 s_2^2 (c_2^2 + c_4 - s_2^2) + s_4 c_2^2) - \\ & L F_{1,0}[m_1] (g_1^2 + g_2^2)^2) \frac{1}{96} s_2^2 s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (3 \sqrt{p} \sqrt{p} \sqrt{p} (c_2^2 + 2 c_2^2) (g_1^2 + 3 g_2^2) - \\ & 3 \sqrt{p} \sqrt{p} \sqrt{p} (c_2^2 - 2 s_2^2) - \sum c_2^2 (g_1^2 + g_2^2) + g_1^2 (1 + 4 c_2^2)) L F_{2,0}[m_1] + \\ & \frac{1}{8} c_2^2 (18 c_2^2 \sqrt{p} \sqrt{p} (g_1^4 + 6 g_1^2 g_2^2 + 16 c_2^2) + 9 g_1^2 (1 + 4 c_2^2) + \\ & 16 s_4^2 \sqrt{p} \sqrt{p} [g_1^4 - 6 g_1^2 g_2^2 + 16 c_2^2] + 9 g_1^2 (1 - 4 c_2^2) - \\ & \sum c_2^2 (g_1^2 + 9 g_1^2 g_2^2 - 8 + 3 c_2^2 (1 + 6 c_2^2)) + 18 g_1^2 g_2^2 (-4 - 9 c_2^2)) L F_{3,0}[m_1] + \\ & \frac{1}{32} c_2^2 (g_1^2 + g_2^2) (24 c_2^2 - (c_2^2 \sqrt{p} \sqrt{p} + s_2^2 \sqrt{p} \sqrt{p}) - \sum c_2^2 (-5 - 6 c_2^2)) - \\ & L F_{4,1}[m_1] + \frac{1}{15} \sum c_2^2 g_2^2 (g_1^2 + g_2^2) L F_{5,2}[m_1] - \\ & \frac{1}{64} \sum s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (4 s_2^2 (c_2^2 + c_4 - s_2^2) + s_4 c_2^2) L F_{1,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{24} \sum s_2^2 s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} (g_1^2 + g_2^2) L F_{2,0}[m_1] + \frac{1}{24} \sum c_2^2 g_1^2 L F_{3,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{64} s_4^4 s_2^2 s_4^2 \frac{1}{m_1^2} \sqrt{p} \sqrt{p} (2 s_2^2 (c_2^2 + c_4 - s_2^2) + s_4 c_2^2) L F_{1,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{8} s_2^2 s_4^2 s_2^2 s_4^2 \frac{1}{m_1^2} \sqrt{p} \sqrt{p} (g_1^2 + g_2^2) (c_2^2 - 2 s_2^2) L F_{2,0}[m_1] + \\ & \frac{1}{256} s_4^4 s_2^2 c_2^2 \sqrt{p} \sqrt{p} L F_{3,0}[m_1] + \frac{1}{256} s_4^4 s_2^2 c_2^2 (g_1^2 - (-1 - 9 c_2^2 + 6 c_4 - 6 s_2^2) + 3 g_2^2 (-1 + 3 c_2^2 + 2 c_4 - 2 s_2^2)) \\ & L F_{1,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{256} s_4^4 s_2^2 (g_1^2 (1 + 7 c_4 - 10 s_2^2) + g_2^2 (9 + 7 c_4 - 10 s_2^2)) L F_{1,1}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{256} s_4^4 s_2^2 (g_1^2 (-1 - 15 c_2^2) + 3 g_2^2 (-1 + 5 c_2^2)) L F_{1,2}[m_1] (g_1^2 + g_2^2)^2 - \\ & \frac{1}{256} s_4^4 s_2^2 (g_1^2 (3 - 5 c_4 - c_2^2) + g_2^2 (1 + 5 c_4 - 2 s_2^2)) L F_{2,0}[m_1] (g_1^2 + g_2^2)^2 + \\ & \frac{1}{16} s_4^4 s_2^2 (g_1^2 (-3 + 11 c_4 - 8 s_2^2) + g_2^2 (1 + 11 c_4 - 8 s_2^2)) L F_{2,1}[m_1] (g_1^2 + g_2^2)^2 + \frac{1}{4688} \\ & (-9 g_1^2 g_2^2 (-1 + c_4) (3 - 2 c_2^2 (5 + 3 c_4) + c_4 (8 + 9 c_4)) + 24 s_2^2 (g_1^2 + 2 s_2^2 + c_4^2) - \\ & 3 g_1^2 (1 + 6 c_2^2 (-1 + c_4) + 2 c_2^2 s_2^2) + 3 c_4 (1 + 3 c_4 + 3 c_4^2 + 3 s_6^4) - \\ & g_2^2 (2 c_2^2 (17 + 9 c_4 (5 + 6 c_4) + 36 s_2^2) + 9 (-9 c_4 (9 + 3 c_4 (-1 + c_4) + 8 s_2^4)) - \\ & g_1^2 g_2^2 (2 c_2^2 (-37 - 9 c_4 (10 + 3 c_4) - 108 s_2^2) + \\ & 9 (9 (3 c_4 + 3 c_4 (-5 + 9 c_4) + 24 s_2^2)) L F_{3,0}[m_1] + \\ & \frac{1}{16} c_2^2 (g_1^2 - 13 + 3 c_4 (6 c_4 + 12 s_2^2)) L F_{4,1}[m_1] + \\ & \frac{1}{64} s_4^4 c_2^2 (g_1^2 + g_2^2) L F_{5,2}[m_1] + \frac{1}{36} s_4^4 c_2^2 (g_1^2 + g_2^2) L F_{5,2}[m_1] + \\ & \frac{1}{24} s_4^4 c_2^2 (g_1^2 + g_2^2) L F_{4,1}[\tilde{\mu}] - \frac{1}{2} g_1^4 c_2^2 (g_1^2 + g_2^2) L F_{5,2}[\tilde{\mu}] - \\ & \frac{1}{64} \frac{1}{m_1^2} c_2^2 L F_{1$$